

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 3: LECTURE NOTE

DIFFERENTIATION

The Derivative of a Function

Definition 1 (Definition of derivative). *The derivative of f at x is*

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

*provided the limit exists. For all x for which this limit exists, $f'(x)$ is a function of x . The process of finding the derivative of a function is called **differentiation**. In addition to $f'(x)$, other notations are used to denote the derivative of $y = f(x)$. The most common are*

$$f'(x), \quad \frac{dy}{dx}, \quad y', \quad \frac{d}{dx}[f(x)], \quad D_x[y]$$

The notation $\frac{dy}{dx}$ is read "the derivative of y with respect to x or simply "dy, dx".

Example 2. Find the derivative of $f(x) = x^3 + 2x$ by the limit process.

Solution.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 + 2(x+h)] - [x^3 + 2x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{[x^3 + 3x^2h + 3xh^2 + h^3 + 2x + 2h] - [x^3 + 2x]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 + 2h}{h} \\ &= \lim_{h \rightarrow 0} [3x^2 + 3xh + h^2 + 2] \\ &= 3x^2 + 2 \end{aligned}$$

Example 3. Find $\frac{dy}{dt}$ of $y = \frac{2}{t}$ by the limit process.

Solution. Considering $y = f(t)$, we obtain

$\frac{dy}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$	Definition of derivative
$= \lim_{h \rightarrow 0} \frac{\frac{2}{t+h} - \frac{2}{t}}{h}$	$f(t+h) = \frac{2}{t+h}$ and $f(t) = \frac{2}{t}$
$= \lim_{h \rightarrow 0} \frac{\frac{2t - 2(t+h)}{t(t+h)}}{h}$	Combine fractions in numerator
$= \lim_{h \rightarrow 0} \frac{-2h}{t(t+h)h}$	Divide out common factor
$= \lim_{h \rightarrow 0} \frac{-2}{t(t+h)}$	Simplify
$= -\frac{2}{t^2}$	Evaluate limit as $h \rightarrow 0$

Example 4 (The derivative of the sine function). *Show that*

$$\frac{d}{dx}[\sin x] = \cos x.$$

Proof. *Applying the limit process:*

$$\begin{aligned} \frac{d}{dx}[\sin x] &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} && \text{Definition of derivative} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} && \sin(\theta + \phi) = \sin \theta \cos \phi + \sin \phi \cos \theta \\ &= \lim_{h \rightarrow 0} \left[\sin x \frac{\cos h - 1}{h} + \cos x \frac{\sin h}{h} \right] \\ &= \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= \cos x && \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0, \quad \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \end{aligned}$$

Example 5 (The derivative of the cosine function). *Show that*

$$\frac{d}{dx}[\cos x] = -\sin x.$$

Proof. . Applying the limit process:

$$\begin{aligned} \frac{d}{dx}[\cos x] &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} && \text{Definition of derivative} \\ &= \lim_{h \rightarrow 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h} && \cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi \\ &= \lim_{h \rightarrow 0} \left[\cos x \frac{\cos h - 1}{h} - \sin x \frac{\sin h}{h} \right] \\ &= \cos x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \sin x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= -\sin x && \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0, \quad \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \end{aligned}$$

□

Theorem 6 (Differentiability Implies Continuity). *If f is differentiable at $x = c$, then f is continuous at $x = c$.*

Proof. We can prove that f is continuous at $x = c$ by showing that $f(x)$ approaches $f(c)$ as $x \rightarrow c$. To do this, use the differentiability of f at $x = c$, and consider the following limit.

$$\begin{aligned} \lim_{x \rightarrow c} [f(x) - f(c)] &= \lim_{x \rightarrow c} \frac{x - c}{x - c} [f(x) - f(c)] \\ &= \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} \right] [x - c] \\ &= \lim_{x \rightarrow c} \left[\frac{f(x) - f(c)}{x - c} \right] \cdot \lim_{x \rightarrow c} [x - c] \\ &= f'(c) \cdot 0 = 0 \end{aligned}$$

Because the difference $f(x) - f(c)$ approaches zero as $x \rightarrow c$, we can conclude that

$$\lim_{x \rightarrow c} f(x) = f(c).$$

So, f is continuous at $x = c$.

□

Basic Differentiation Rules

Theorem 7 (The Constant Rule). *The derivative of a constant function is 0. That is, if c is a real number, then*

$$\boxed{\frac{d}{dx}[c] = 0}. \quad (1)$$

Proof. Let $f(x) = c$. Then, by the limit definition of the derivative,

$$\frac{d}{dx}[c] = \frac{d}{dx}[f(x)] = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0.$$

Theorem 8. *The derivative of the identity function $f(x) = x$ is 1. That is,*

$$\boxed{\frac{d}{dx}[x] = 1}. \quad (2)$$

Proof. By the limit definition of the derivative,

$$\begin{aligned} \frac{d}{dx}[x] &= \frac{d}{dx}[f(x)] \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} \\ &= \lim_{h \rightarrow 0} 1 \\ &= 1. \end{aligned}$$

Theorem 9 (The Power Rule). *If n is a positive integer, then the function $f(x) = x^n$ is differentiable and*

$$\boxed{\frac{d}{dx}[x^n] = nx^{n-1}}.$$

Proof. For a positive integer n , let $f(x) = x^n$. Then the binomial expansion produces,

$$\begin{aligned} \frac{d}{dx}[x^n] &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{[x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots + \binom{n}{n-1}xh^{n-1} + h^n] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \dots + \binom{n}{n-1}xh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} [nx^{n-1} + \binom{n}{2}x^{n-2}h + \dots + nxh^{n-2} + h^{n-1}] \\ &= nx^{n-1} + 0 + \dots + 0 + 0 \\ &= nx^{n-1} \end{aligned}$$

□

Theorem 10 (Derivative of Square Root Function). *The derivative of $f(x) = \sqrt{x}$ is $\frac{1}{2\sqrt{x}}$. That is,*

$$\boxed{\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}}}. \quad (3)$$

Proof. . For every $x > 0$, by the limit definition of the derivative,

$$\begin{aligned} \frac{d}{dx}[\sqrt{x}] &= \frac{d}{dx}[f(x)] \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{[\sqrt{x+h} - \sqrt{x}][\sqrt{x+h} + \sqrt{x}]}{h[\sqrt{x+h} + \sqrt{x}]} \\ &= \lim_{h \rightarrow 0} \frac{[x+h] - x}{h[\sqrt{x+h} + \sqrt{x}]} \\ &= \lim_{h \rightarrow 0} \frac{+h}{h[\sqrt{x+h} + \sqrt{x}]} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

□

Example 11. *Find the derivative of the function.*

(a) $f(x) = x^3$ (b) $f(x) = -3$ (c) $y = 7$ (d) $f(x) = \sqrt[3]{x}$ (e) $y = x^{12}$

(a) *Using the power Rule*

$$\frac{d}{dx}[x^3] = 3x^2.$$

(b) *Using the Constant Rule*

$$\frac{d}{dx}[-3] = 0.$$

(c) *Using the Constant Rule*

$$\frac{dy}{dx} = \frac{d}{dx}[7] = 0.$$

(e) *Using the power Rule*

$$\frac{dy}{dx} = \frac{d}{dx}[x^{12}] = 12x^{11}.$$

Theorem 12 (The Constant Multiple Rule). *If f is a differentiable function and c is a real number, then cf is also differentiable and*

$$\boxed{\frac{d}{dx}[cf(x)] = cf'(x)}.$$

Proof.

$$\begin{aligned}
 \frac{d}{dx}[cf(x)] &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\
 &= \lim_{h \rightarrow 0} c \frac{f(x+h) - f(x)}{h} \\
 &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= cf'(x).
 \end{aligned}$$

□

Example 13. Using the Constant Multiple Rule

$$(a) \quad \frac{d}{dx}[5x^3] = 5 \frac{d}{dx}[x^3] = 5(3x^2) = 15x^2.$$

$$(b) \quad \frac{d}{dx}[-3 \sin x] = -3 \frac{d}{dx}[\sin x] = -3 \cos x.$$

$$(c) \quad \frac{d}{dx}[2\sqrt{x}] = 2 \frac{d}{dx}[\sqrt{x}] = 2\left(\frac{1}{2\sqrt{x}}\right) = \frac{1}{\sqrt{x}}.$$

$$(d) \quad \frac{d}{dx}\left[-\frac{3}{2}x\right] = -\frac{3}{2} \frac{d}{dx}[x] = -\frac{3}{2}.$$

Remark 14. The Constant Multiple Rule and the Power Rule can be combined into one rule. The combination rule is

$$\boxed{\frac{d}{dx}[cx^n] = cnx^{n-1}.}$$

Theorem 15 (The Sum and Difference Rules). The sum (or difference) of two differentiable functions f and g is itself differentiable. Moreover, the derivative of $f+g$ (or $f-g$) is the sum (or difference) of the derivatives of f and g .

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x) \quad \text{[Sum Rule]} \quad (4)$$

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x) \quad \text{[Difference Rule]} \quad (5)$$

Proof. A proof of the Sum Rule follows from the Sum Rule for limit. (The Difference Rule can be proved in a similar way.)

$$\begin{aligned}
 \frac{d}{dx}[f(x) + g(x)] &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\
 &= \lim_{h \rightarrow 0} \left(\frac{[f(x+h) - f(x)] + [g(x+h) - g(x)]}{h} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right) \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= f'(x) + g'(x)
 \end{aligned}$$

□

Remark 16. The Sum and Difference Rules can be extended to any finite number of functions.

Example 17. Using The Sum and Difference Rules

$$(a) \quad \frac{d}{dx}[x^3 - 4x + 5] = \frac{d}{dx}[x^3] - \frac{d}{dx}[4x] + \frac{d}{dx}[5] = 3x^2 - 4 + 0 = 3x^2 - 4.$$

$$(b) \quad \frac{d}{dx}[x^2 - \sin x] = \frac{d}{dx}[x^2] - \frac{d}{dx}[\sin x] = 2x - (\cos x) = 2x - \cos x.$$

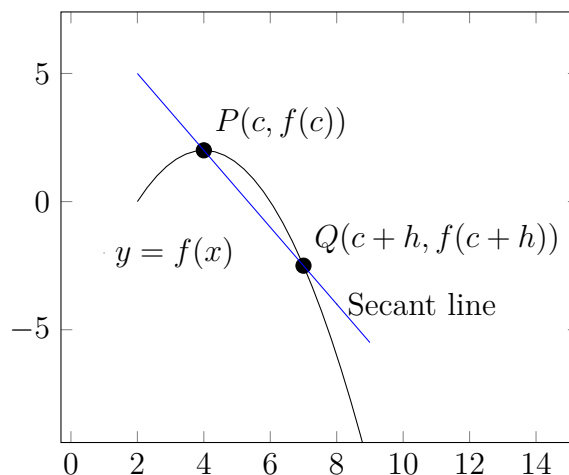
Rates of Change

Suppose a curve (graph of a function f) is given by equation $y = f(x)$. To find the tangent line to the curve at a point $P(c, f(c))$, we consider a near by point $Q(c + h, f(c + h))$. The slope of the secant line through the two points is given by substitution into the slope formula

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad \begin{array}{l} \text{Change in } y \\ \text{Change in } x \end{array}$$

$$m_{PQ} = \frac{f(c+h) - f(c)}{(c+h) - c}$$

$$m_{PQ} = \frac{f(c+h) - f(c)}{h}$$



If we let h approaches 0, then the point Q slides along the curve to the point P and the secant line becomes the tangent line. Thus, we have the following definition.

Definition 18. If f is defined on an open interval containing c , and if the limit

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = m \quad (6)$$

exists, then the line passing through $(c, f(c))$ with slope m is the **tangent line to the graph of f at the point $(c, f(c))$** ; that is, the slope of the tangent line is

$$m = f'(c).$$

The slope of the tangent line to the graph of f at $(c, f(c))$ is also called the **slope of the graph of f at $x = c$** .

Example 19. Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Solution. Since $f'(x) = 2x$ for every x , the slope of the graph at $P(1, 1)$ is given by

$$m = f'(1) = 2.$$

Because the tangent line passes through the point $P(1, 1)$, by the point-slope form of equation of a line formula we get

$$y - 1 = 2(x - 1) \quad \text{or} \quad \boxed{y = 2x - 1}.$$

The derivative can also be used to determine the rate of change of one variable with respect to another. Applications involving rates of change, sometimes referred to as **instantaneous rates of change**, occur in a wide variety of fields. A few examples are population growth rates, production rates, water flow rates, velocity, and acceleration.

A common use for rate of change is to describe the motion of an object moving in a straight line. The function s that gives the position (relative to the origin) of an object as

a function of time t is called a position function. If, over a period of time Δt , the object changes its position by the amount

$$\Delta s = s(t + \Delta t) - s(t)$$

then, by the familiar formula

$$\text{Rate} = \frac{\text{distance}}{\text{time}}$$

the average velocity is

$$\frac{\text{Change in Position}}{\text{Change in time}} = \frac{\Delta s}{\Delta t}.$$

In general, if $s(t)$ is the position function for an object moving along a straight line, then the velocity of the object at time t is

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t} = s'(t) = \frac{d}{dt}[s(t)].$$

The position of a **free-falling object (neglecting air resistance)** under the influence of gravity can be represented by the equation

$$s(t) = \frac{1}{2}gt^2 + v_0t + s_0, \quad \text{Position function}$$

where s_0 is the initial height of the object, v_0 is the initial velocity of the object, and g is the acceleration due to gravity. On Earth, the value of g is approximately $-32 \frac{ft}{s^2}$ (or $-9.8 \frac{m}{s^2}$).

Example 20 (Using the Derivative to Find Velocity). *At time $t = 0$ a diver jumps from a platform diving board that is 32 ft above the water. Because the initial velocity of the diver is $16 \frac{ft}{s}$, the position of the diver is*

$$s(t) = -16t^2 + 16t + 32, \quad \text{Position function}$$

where s is measured in feet and t is measured in seconds.

(a) *When does the diver hit the water?*

To find the time when the diver hits the water, let $s = 0$ and solve for t .

$$\begin{array}{ll} -16t^2 + 16t + 32 = 0 & \text{Set position function equal to } 0. \\ -16(t+1)(t-2) = 0 & \text{Factor.} \\ t = -1, \quad t = 2, & \text{Solve for } t \end{array}$$

Because $t \geq 0$, choose the positive value to conclude that the diver hits the water at $t = 2$ seconds.

(b) *What is the divers velocity at impact?*

The velocity at time t is given by the derivative

$$s'(t) = -32t + 16. \quad \text{Velocity function}$$

So, the velocity at time $t = 2$ is

$$s'(2) = -32(2) + 16 = -48 \text{ ft/s.}$$